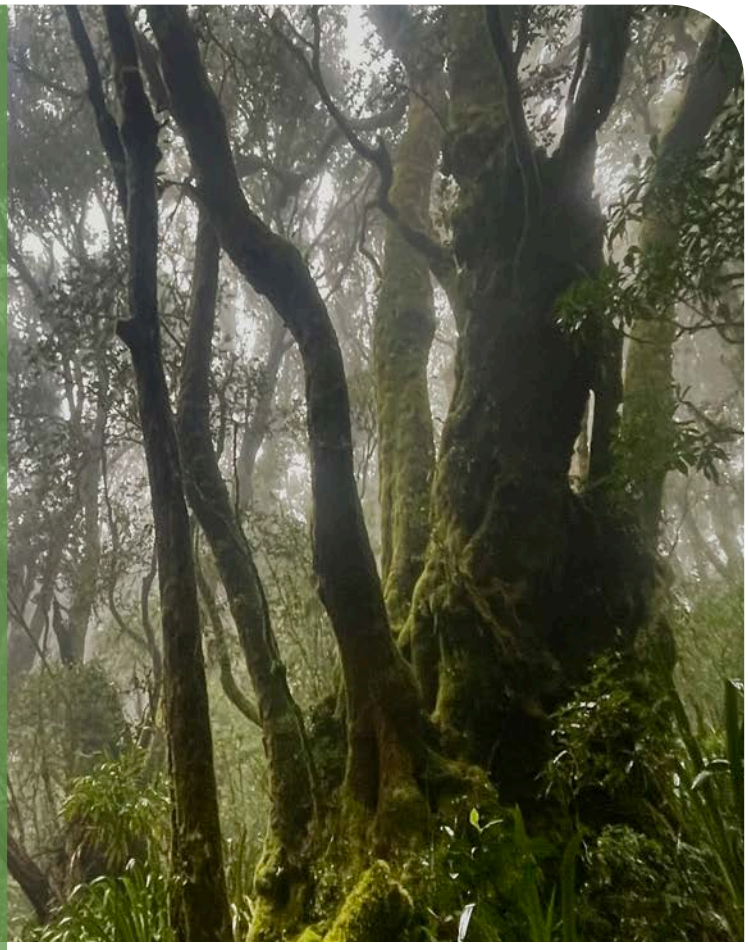
Forests 2030



Protecting & restoring ecosystem integrity

Report by Cyril Kormos [1], Professor Brendan Mackey [2], Andrew Buckwell [2], Virginia Young [2], Sonia Hugh [2], & Dr Pat Norman [2].

1. Wild Heritage; 2. Griffith University

1. Introduction

The world's forests – from lush tropical forests to the boreal forests in the far north, and the temperate forests in between – are all in rapid decline.

Yet forests matter more than ever. They provide life-sustaining ecosystem services, such as clean freshwater, are essential for combating the twin global crises of biodiversity loss and climate change, are critical for maintaining cultural values and upholding Indigenous Peoples' rights (see figures 1 and 2), and help prevent pathogen spillover and pandemics. The fate of the world's forests affects every living thing on the planet.

Not all forests are equal: the critical importance of 'ecosystem integrity'

Not all forests are equal – they can be graded by their degree of 'naturalness', or ecological integrity.

This feature of forests is too often lost in debates about forests. Some forests are not natural but are actually planted 'tree crops', such as monocultures planted for bioenergy or pulp. And many natural forests are in a degraded state. Only about a quarter of the world's forests remain in good ecological condition, where most, if not all, of their native species are present, and the forest structure is largely free of industrial impacts, such as logging and mining. These forests are uniquely important as they provide far more benefits than degraded forests or plantations, and if lost cannot be recovered in



Figure 1: The Kayapo Indigenous Territories protect nearly 11 million hectares of primary rainforest. Photo: Martin Schoeller for the Primary Forests and Climate project. The area protected can be defined by satellite imagery (see Figure 2). Photo: Martin Schoeller for the Primary Forests and Climate project.

timeframes that matter for solving our climate and biodiversity challenges.

'Primary forest' encompasses other commonly used descriptors for forests that sit towards one end of a spectrum of forest ecological condition that at the opposite end reflects increased impact of modern human activities, including commercial logging, infrastructure development, ranching, and mining [1]. Primary forest, therefore, is naturally regenerating forest of native tree species, whose structure, composition and dynamics are dominated by ecological and evolutionary processes.

This definition does not mean that primary forests are uninhabited by humans. On the contrary, the world's primary forests are the customary homelands of Indigenous Peoples who continue to play a critical role in their protection and conservation management [2]. Prior significant human intervention may also have occurred, but was long enough ago to have allowed an ecologically mature forest ecosystem to re-establish. As used here, primary forest encompasses related terms including stable [3], old-growth, mature and virgin forest [4].

We should not talk about forests in generic terms, as if all types of forests are interchangeable and provide the same benefits. We must prioritise the protection of those forests that are in good ecological condition – forests with a high level of ecosystem integrity, such as primary, old growth, and mature forests. This is the most efficient and cost-effective way to maximise the vital benefits forests provide, including biodoversity, carbon storage, and freshwater flows. This is already mandated in international forest policy. Our second priority should be to restore forests that have been degraded so they can recover their ecosystem integrity over time.

Therefore, ending the global forest crisis means ending two threatening processes:

- **deforestation** - the permanent and complete clearing of forest and its replacement with another land use, such as agriculture; and
- forest **degradation** - the clearing or damaging forest, but without permanent conversion to another land use.

The purpose of this report is to help explain why forest policies for 2030 and beyond must embrace ecosystem integrity as a core guiding principle, particularly with respect to forests and climate change.

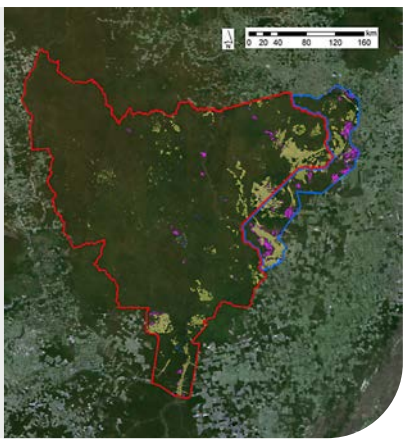


Figure 2: Primary forests provide an important role in upholding Indigenous Peoples' rights. The Kayapo Indigenous Territories are so well defended against unlawful activities that the outline is visible from space.

The urgency of action

We have lost one third of the world's forests – or around 2 billion hectares since 1950 [5]. This is about as much forest loss over a 75-year period as was lost in the preceding 10,000 years (figure 3) [6]. We have also degraded hundreds of millions of hectares of forest so that today only about 25-30% of the world's remaining forests are primary, old growth or mature forests [7]. This massive deforestation and forest degradation has had catastrophic impacts on forest carbon stocks and biodiversity, it has been a major contributing factor to global freshwater and desertification crises and has had enormous impacts on the wellbeing and rights of Indigenous Peoples and local communities.

The United Nations is fully aware of the magnitude and severity of the forest crisis - and has a solution: the **United Nations Strategic Plan for Forests 2017-2030** calls for an end to deforestation and forest degradation by 2030. The UN target for ending deforestation and forest degradation by 2030 was reiterated in the **Glasgow Leader's Declaration on Forests and Land Use** in 2021, and by **UN Framework Convention on Climate Change Global Stocktake Decision** in 2023 (1/CMA.5) and is echoed by the **Kunming-Montreal Global Biodiversity's Framework**'s emphasis on ecosystem integrity (Goal A, Targets 1 and 2).

The **UNEP 2025 Emissions Gap Report** noted that "despite the key role of fossil fuels in driving total emissions, deforestation and land-use change was decisive for the rapid increase in 2024 emissions – increasing by 21% in 2024 and responsible for 53% of the overall increase in global GHG emissions".

We now have 4 years to achieve this critical objective!

We've lost 1/3 of the world's forests – or about 2 billion hectares, since the start of agriculture roughly 10,000 years ago

Roughly half of that loss (1 billion hectares) occurred before 1950. The next billion hectares has been cleared since 1950.

Only about 25% of our remaining forests are primary forests.

We've degraded at least 400 million hectares since 1990.

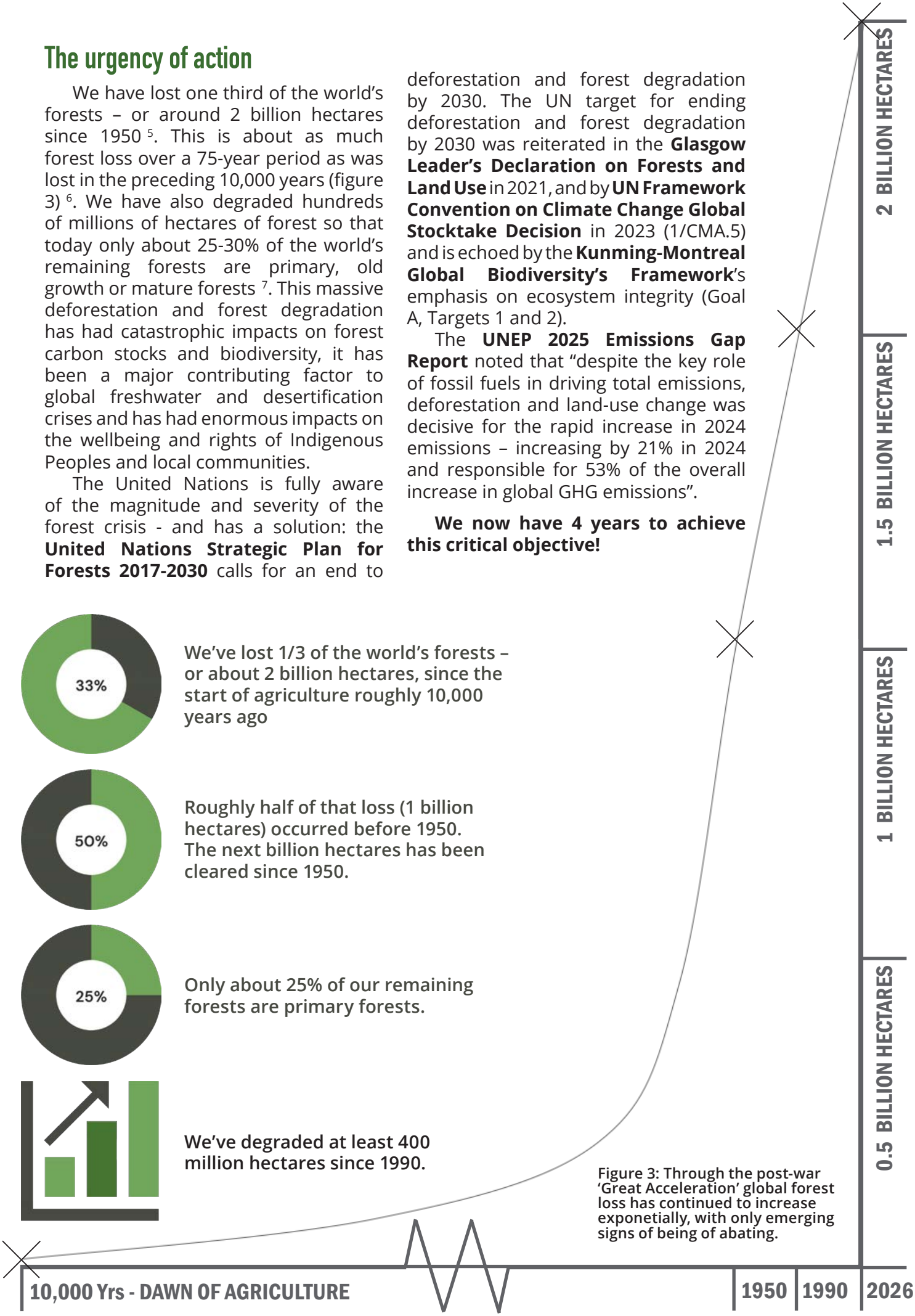


Figure 3: Through the post-war 'Great Acceleration' global forest loss has continued to increase exponetially, with only emerging signs of being of abating.

2. What is ECOSYSTEM INTEGRITY

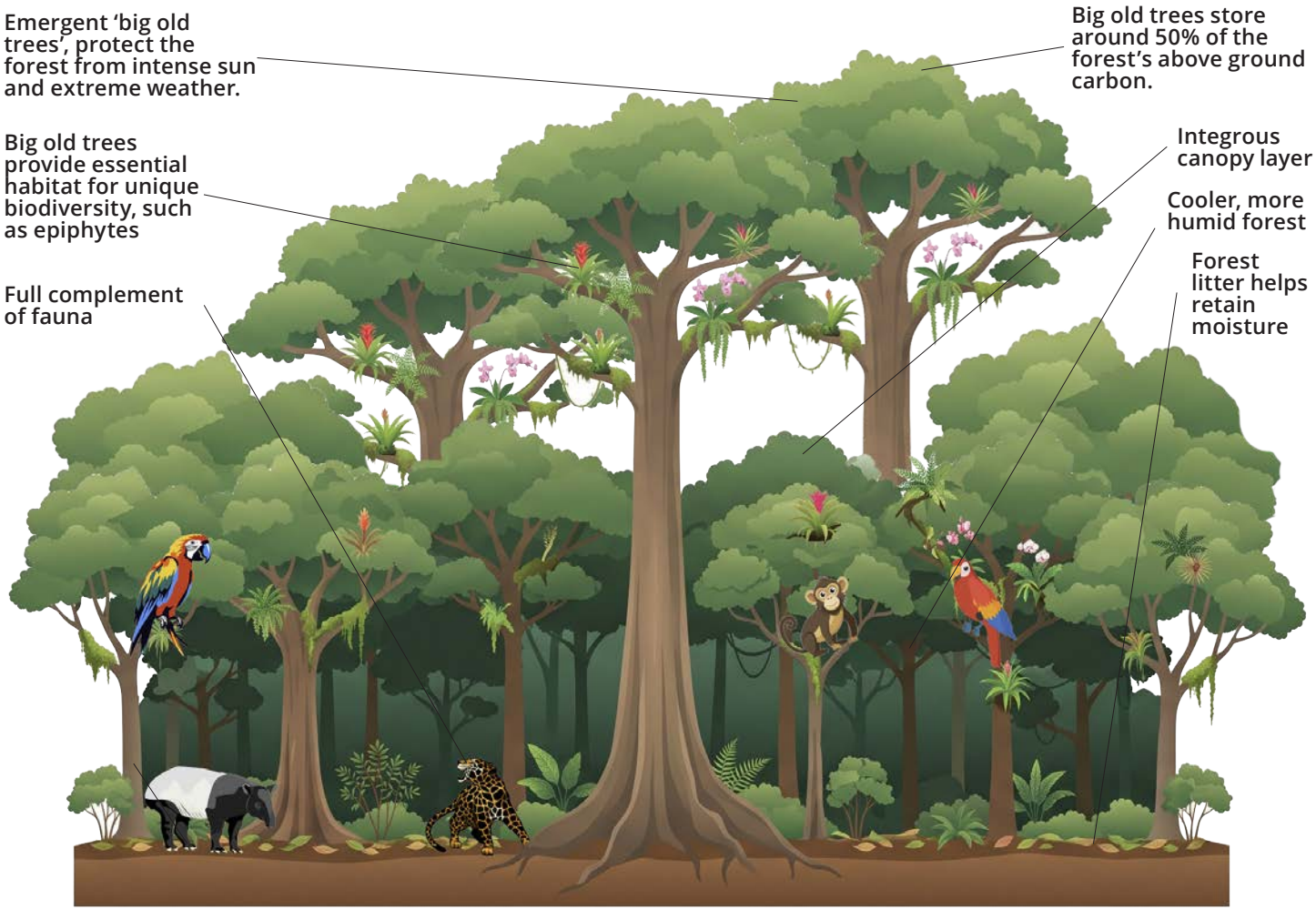
Defining ecosystem integrity

Ecosystem integrity is a critical concept but poorly understood in policy circles. Ecosystem integrity is a measure of ecosystem wholeness. High ecosystem integrity means an ecosystem has not been significantly modified by industrialized activities and still retains most of, if not all, of its native **biodiversity, vegetation structure,** and **functions** (figure 4) [8].

A natural complement of biodiversity helps forests both resist and adapt to stressors that are increasing with climate change. Forests need healthy populations of their natural species because every species native to a specific forest ecosystem plays a role in the forest [9]. For example, birds, mammals and insects disperse seeds or pollinate plants or trees. Fungi break down and decompose dead vegetation, cycle nutrients, and make soils more absorptive for water [10] and for storing carbon. As species are increasingly removed, or species populations are reduced, the forest's ability to maintain key ecological processes also decreases, and the forest becomes weaker and less resilient [11]. Loss of biodiversity results in forest degradation, putting key forest functions, such as carbon storage or freshwater flows, at risk.

Retaining forest structure, especially a forest's big, old trees, is essential. Big trees play a central role in protecting the forest from excessive sunlight, wind and storms [12]. They have strong genes, so it is important to forest health that they continue to reproduce. They also provide habitat for dozens of species [13] and store most of the forest's vegetation carbon. Removing big trees greatly weakens a forest [14].

Figure 4: Features of a primary rainforest with a high level of ecosystem integrity.



Why Ecosystem integrity is important

A forest with high ecosystem integrity, such as a primary or old growth forest, has been evolving for thousands, or even millions of years. As a result, it is extremely well adapted to local conditions and, having persisted through natural climatic shifts, has a greater capacity to adapt to future change. It is strong, and more stable than planted and other young forests, and therefore *maximises* forest benefits.

A forest with a high level of ecosystem integrity is not only better able to resist disturbances (e.g., fire, extreme weather, disease, insect outbreaks), but is also more resilient; i.e. it is quicker to recover when change or disturbance occurs, so it is better able to persist for the long term – up to millions of years [15].

A forest with high ecosystem integrity also provides more benefits than a degraded forest:

- more different types of ecosystem services (biodiversity, erosion control, pollination, freshwater quality, fire resistance etc.);
- a greater quantity of each ecosystem service (more carbon per hectare, more species, more freshwater retained in the ecosystem etc.); and
- higher quality ecosystem services (cleaner freshwater, and lower risk carbon stocks) [16].

These greater ecosystem service benefits are illustrated below with respect to biodiversity and carbon though there are many other important examples, such as greater fire resistance, freshwater quality and flows, erosion control, and pollination.



Giant Sequoia (*Sequoiadendron giganteum*) in Sierra Neveda, California, USA. Source: A. Buckwell.

3. Primary Forests & BIODIVERISTY

- Primary forests are the most biodiverse terrestrial ecosystems on Earth. They are home to **more species** than a degraded forest of the same type (and far more species than a planted forest or commercial plantation) [17].
- Primary forests also have **more unique biodiversity:** more unique species 'assemblages', i.e., groups of species, more endemic species (species not found elsewhere), and more rare or threatened species [18].
- Tropical forests are home to at least two thirds of the world's terrestrial species [19]. Some estimates suggest forests could house as much as 80% of terrestrial species [20]. They are essential to protecting the diversity of life on Earth.



Primary tropical forests have extraordinary biodiversity, including this Toco Toucan from South America. Source: Matheus Figueiredo from Pexels.com

4. FOREST Degradation & Deforestation

Deforestation & forest degradation are related but different concepts.

Deforestation means removing at least 90% of forest cover in a given area **and** *permanently* converting the cleared land to another land use (figure 5, right), such as agriculture or urban development (figure 6, right). **Forest degradation** is often more subtle [21]. Generally speaking, some forest cover remains, but the forest's structure, composition, and ecological functions have been reduced, depleted, and downgraded through logging, fragmentation from roads, overhunting, or some other impact. As a result, forest degradation is often overlooked, despite affecting vast areas globally (figure 7, right; figure 8 over the page) – a far larger area than is deforested annually.

However, technically speaking, the key difference between the definition of deforestation and degradation is the **permanence of the damage**, rather than the extent of the damage. An area of forest can be clear cut, removing every single tree, but if the intention is to replant, or to allow the forest to regrow, then technically speaking there has not been any deforestation – only degradation. Thus, clear cutting an area of natural forest in Europe or North America which will be replanted with a monoculture of cloned seedlings does *not* count as deforestation. However, clear cutting an area of forest in the Amazon, and replacing it with soy or cattle *does* count as deforestation. In other words, it is not always possible to tell just by looking at a picture of cleared forest land whether that area is deforestation or forest degradation.

Thus, clear cutting an area of natural forest in Europe or



Figure 5: High integrity primary forest in the Amazon. Source: Apple Maps



Figure 6: Deforestation in the Amazon for cattle grazing. Source: Adobe



Figure 7: Degraded tropical forest as a result of selective logging, Solomon Islands. Source: Google Earth

DEFORESTATION

FOREST DEGRDATION

North America, which will be replanted with a monoculture of cloned seedlings does not count as deforestation! However, clear cutting an area of forest in the Amazon and replacing it with soy or cattle does count as deforestation. In other words, it is not always possible to tell just by looking at a picture of cleared forest land whether that area will count as deforestation or degradation.

That said, degradation is usually understood to involve a reduction in tree cover and retreat of animal species which erodes a forest's naturalness, weakening it and reducing its ability to deliver ecosystem services. For example, selective logging reduces ecosystem integrity and primary forest characteristics by removing critically important big, old trees (see feature below) and large hollow-bearing trees - the loss of which cascades through species dependent on them. Retreat of faunal species also has important ecological impacts, for example if that species is a pollinator or seed disperser, or if a food chain is disrupted. Degradation also allows sunlight and wind into the forest, altering microclimates, drying out the forest and making it more fire prone.

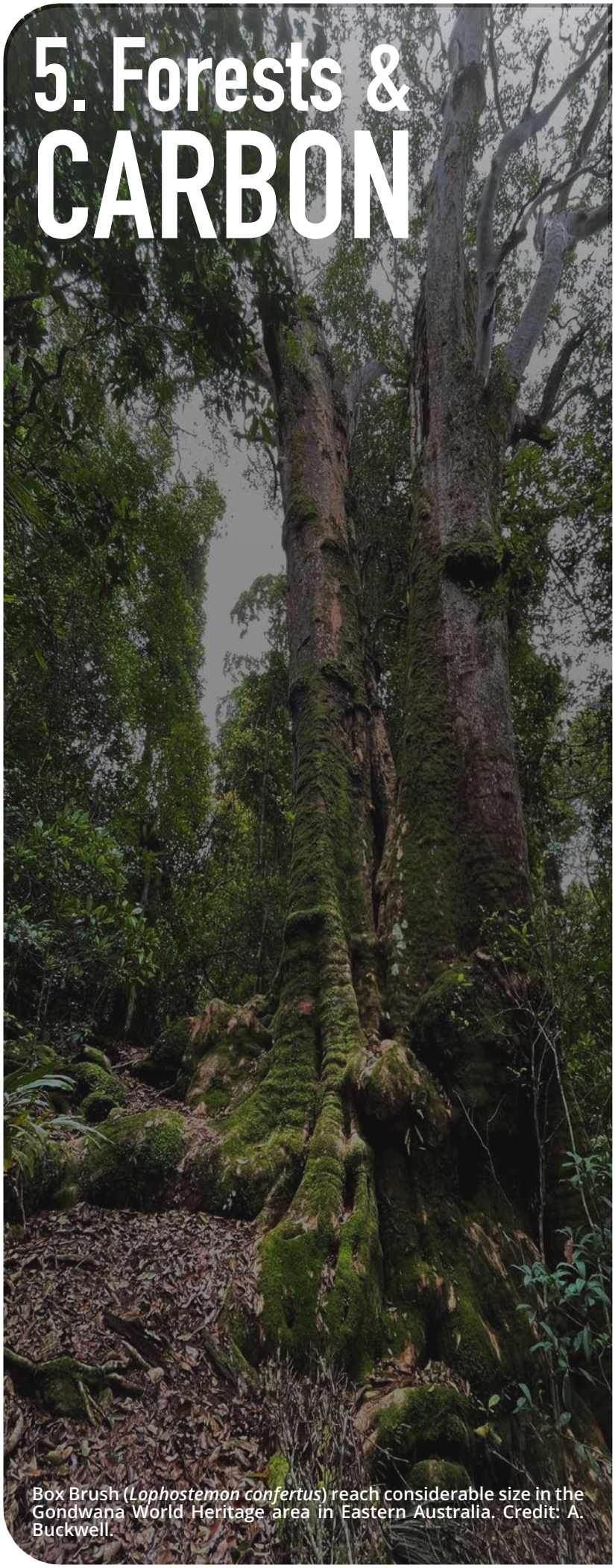
Degradation also reduces forest carbon storage [22]. Primary forests hold far more carbon, more securely, than logged or regrowth forests and the carbon 'debts' from logging of primary forests can take centuries to repay, if it is ever, as the big older trees that are generally targeted that store most of the forest carbon. Primary forests are being lost faster than global statistics suggest as these statistics ignore key differences between planted monocultures and natural forests and between primary and degraded natural forests. Too often the result is a policy focus on maintaining or increasing forest tree cover irrespective of ecological condition, and at the expense of protecting and restoring forest ecosystem integrity

Protecting remaining forests with high ecosystem integrity, rather than relying on forest restoration alone, is essential for safeguarding biodiversity, climate regulation, and climate resilience.

Figure 8: Forest loss in the Canadian 'Wetbelt' forest in the Gordon River Valley, British Columbia. Although this might look like a clear case of 'deforestation', it might not be considered as such in national datasets and in climate policy *if* forest managers intend to allow this clearing to regrow, or have plans to replant seedlings in the future. Source: T J Watt, Wikimedia.



5. Forests & CARBON



Box Brush (*Lophostemon confertus*) reach considerable size in the Gondwana World Heritage area in Eastern Australia. Credit: A. Buckwell.

Key terms & concepts

Above & below ground carbon

All ecosystems absorb or 'sequester' carbon through photosynthesis. As a result, roughly half of all living matter (biomass) in plants and animals is carbon.

Forests store carbon **above ground** in vegetation – in trees, plants, and forest litter (i.e., leaves, branches, dead trees on the forest floor). Most carbon assessments do not include carbon in the wildlife.

Forests also store carbon **below ground**, in root systems, and in soils, through a range of processes, including decomposition of vegetation, root system carbon release, and microbial activity.

Adding up the carbon in vegetation and in roots and soils gives you the **Total Ecosystem Carbon**. Any carbon estimate that does not include both above and below ground carbon understates the amount of carbon in the ecosystem.

Stock, sinks & sources

There is a difference between the carbon stock and the carbon sink, or source.

The carbon **stock** is the total ecosystem carbon – in other words, the carbon that has already been absorbed from the atmosphere by the forest and that has accumulated in the trees, deadwood and soil. The most important climate mitigation value of a primary forest is its ability to absorb, accumulate and retain its existing carbon stock over the long-term.

The carbon **sink** is the amount of net carbon absorbed over a given period of time (typically one year). This varies year to year. If an ecosystem loses more carbon than it absorbs in a given year then that ecosystem can be labelled as a **net source** of carbon dioxide over that time period.

Given the urgency of the climate crisis, the top mitigation priority is keeping carbon dioxide out of the atmosphere for the long term; i.e., at least one hundred years [23]. Carbon stored in a forest for a few decades before being re-emitted to

the atmosphere represents a short-term **carbon flux** and provides little or no mitigation value [24].

Policy that focuses on net annual fluxes of carbon into and out of the atmosphere misses a forest's main mitigation value, which is the ability to retain carbon (i.e. keep it out of the atmosphere) over the long term.

Carbon storage in different types of forest

Forests globally store a lot of carbon. There is as much carbon in the world's forests as there is in the atmosphere. However, different types of forest store carbon differently [25].

For example, **tropical rainforests**, that benefit from abundant sun, warmth, and water, and grow year-round, have high rates of primary production, which produces a large volume of vegetation, and as a result, have a large amount of carbon stored in living trees (figure 10).

However, tropical forests often grow on poor soils, so they often have low soil carbon content – nutrients are quickly absorbed out of soils and into vegetation [26]. Tropical forests have few big old trees per hectare (a few dozen), and carbon in a tropical forest is highly concentrated in these few big trees. The exceptions are tropical forests found on peat formations, which have a high carbon content, such as in the Democratic Republic of Congo or Indonesia. These forests have large amounts of above *and* below ground carbon so their total ecosystem carbon is very high [27].

Temperate (figures 9 and 11) and **boreal** (figure 12) forests are found in colder climates with shorter growing seasons and therefore are not as productive as tropical forests. As a result, above ground carbon in a temperate or boreal forest is lower than in the tropics.

However, these forests have lower decompensation rates, so more carbon is retained in dead wood and the soil [28]. As a result, some of the most carbon dense forest ecosystems on Earth are temperate; for example, mountain ash eucalypt forests in South East Australia and redwood forests in California [29].

Again, when boreal or temperate forests coincide with peat formations (often the case in boreal forests), soil carbon can be higher than above ground carbon.

Carbon longevity

To contribute to solving the climate crisis, carbon needs to be stored in an ecosystem for the long term. We therefore need forests to persist as they have done for millenia. We must retain carbon stocks in our most secure

Figure 9: Temperate forest, such as this European Beech and Oak, have lower growth rates than tropical forests, as they have shorter growing seasons. Above ground carbon here is lower, but these forests have larger dead wood ad soil carbon stocks.



of forests: primary forests and Intact Forest Landscapes [30], and wherever feasible, recover the natural carbon stock in degraded forests.

Storing carbon in forest trees an ecosystem for just a few decades and then logging them to release the carbon back it into the atmosphere is of little, or no, climate benefit [31]. Moreover, young regrowth and degraded forests are more vulnerable to releasing carbon as a result of drought, fire, disease and insect outbreaks.

They are also more likely to be cleared for agriculture [32]. This means the carbon absorbed by a young or degraded forest is at much greater risk of being re-released into the atmosphere after a short period of time. Therefore, a top priority is protecting ecosystems, such as primary and old growth forests, that already store a lot of carbon, and can store this carbon more safely for the long term [33].

However, the high-risk carbon storage in a young or degraded forest can still be important if that forest eventually matures into an old growth forest and begins to provide real climate mitigation benefits. But this higher-risk approach must complement, not replace, a focus on protecting ecosystems with high ecosystem integrity.

This concept of longevity of the carbon storage is a crucial point that is often missed in forest carbon projects. Twenty or thirty years of temporary carbon storage may appear to be the logical planning horizon for a forest restoration project, but it has no meaningful impact on climate change. To illustrate the point, carbon re-emitted into the atmosphere thirty years from now will be emitted just past mid-21st century, by which time we must have already achieved net zero emissions in line with climate targets in the Paris Agreement. We can therefore draw-up the following carbon mitigation hierarchy.

Carbon mitigation hierarchy

Priority 1: Protect primary forests

Primary forests have already accumulated a large carbon stock [34] and provide more and better ecosystem services [35]. Their high ecosystem integrity means they are far more stable and at lower risk of losing their carbon when compared to a degraded forest, planted natural forest, or monoculture plantation. Protecting this large, stable, long-lived, low-risk existing stock is also the most cost-effective way of avoiding emissions and securing carbon [36]. It is therefore the top priority [37].

Priority 2: Allow mature forest to age

The next highest priority is allowing a mature forest to 'ecologically age' and achieve old growth status by protecting it from further logging and degradation [38]. This allows medium-sized trees to grow into big trees, old trees, which will sequester more carbon than a degraded forest comprised of very young trees over the same period of time. Prioritising the restoration of mature forests, sometimes referred to as *pro*forestation, is therefore the next priority.

Priority 3: Reforestation

The third priority is to allow naturally forested land, that has been cleared or degraded, to regrow, or to actively support its restoration if it is too degraded to recover on its own [39]. Natural regeneration is the most cost-effective action, as actively restoring forests can be very expensive and doesn't always succeed [40].

While younger forests absorb carbon at higher rates than a primary forest, it takes decades, or centuries before they have a comparable accumulated carbon stock to that found in a mature or old growth forest. The difference between a fast-growing young forest and a primary forest is similar to the difference between managing a small, risky investment with a potentially high rate of return, and managing a large endowment with a modest, but solid long-term record of returns. A younger forest is a high-risk investment - it may one day provide significant biodiversity, carbon, and other benefits, but during the investment period it may very easily be lost to drought, fire, insect outbreaks, or other threats as it is not nearly as robust as a primary forest. Conversely, an old growth forest may not accumulate much more in the way of carbon or biodiversity – but it already has an exceptionally high value. As with an endowment, the goal isn't rapid growth; it's ensuring that what has already been accumulated is not lost as a result of poor management!

Figure 10: Carbon stocks in primary tropical forests.

Carbon stores in Primary Tropical Forests

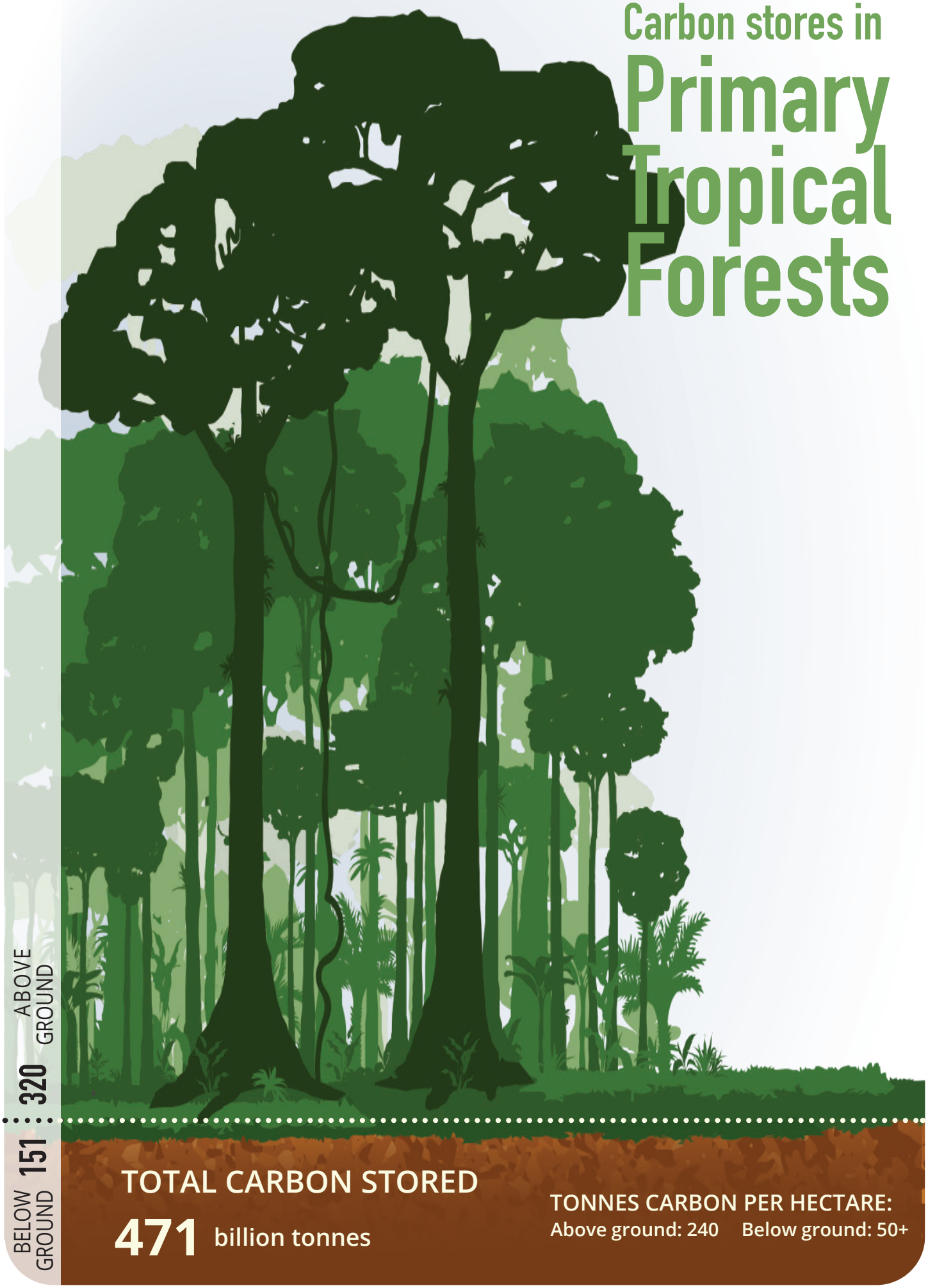


Figure 11: Carbon stocks in primary temperate forests.

Carbon stores in Primary Temperate Forests

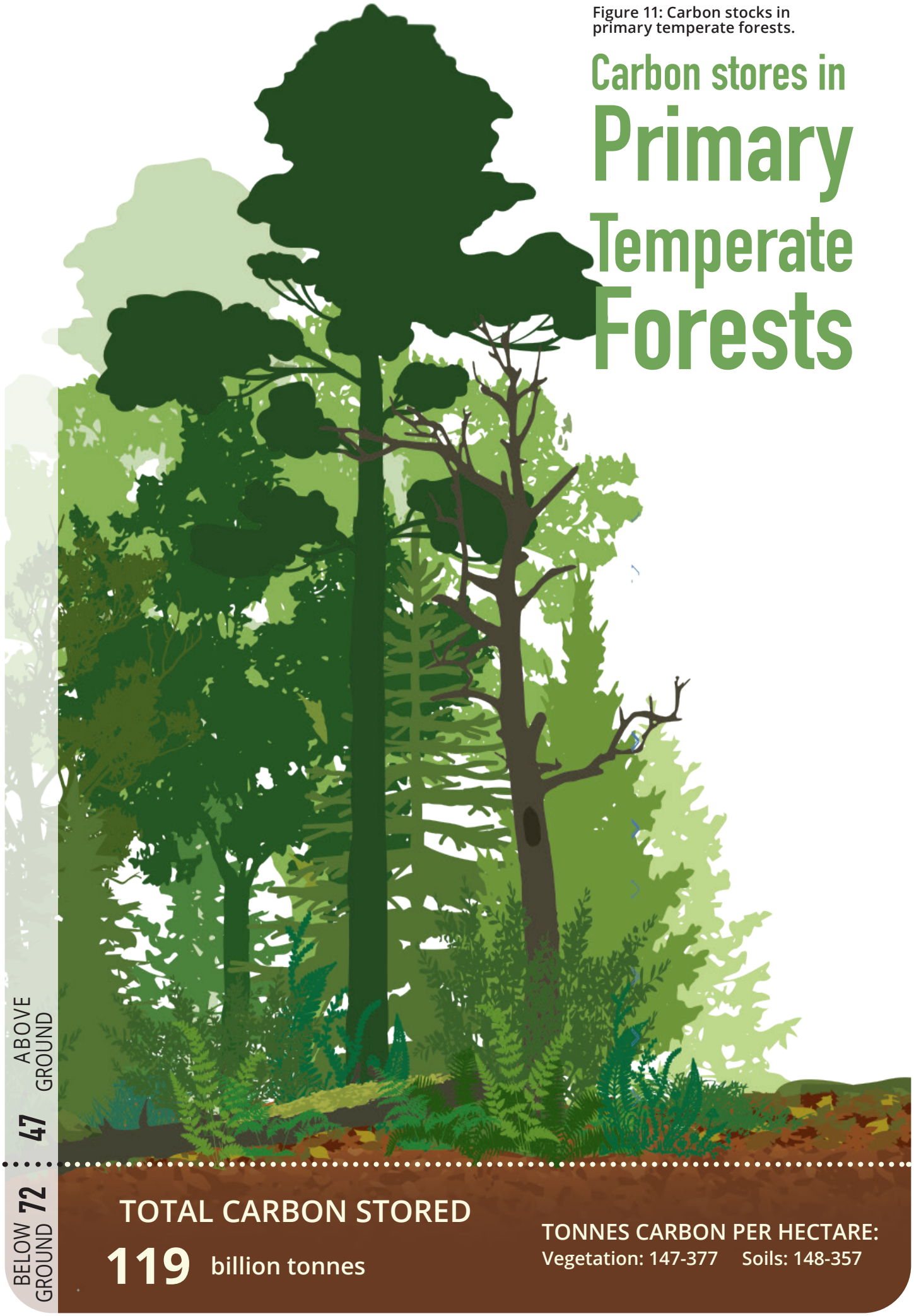


Figure 12: Carbon stocks in primary boreal forests.



The critical importance of BIG OLD TREES

The critical importance of big trees

Trees can live for centuries, or even for thousands of years, in many cases growing very large as they age. In addition to playing a critical role in forest structure, big, old trees (figure 13) are also crucial to carbon storage [41]. Carbon is hyper-concentrated in these trees. In fact, several studies suggest that 1% of the trees globally (i.e. the very biggest trees) store up to 50% of the above ground carbon [42].

Contrary to popular belief, big, old trees are not trees in decline. They actually absorb more carbon than younger trees, as they have more and bigger leaves (or pine needles), larger crowns, more surface area on their trunks, and more extensive root networks. They continue to absorb large amounts of carbon dioxide throughout their very long lifespans [43].

Therefore, removing even a few big, old trees per hectare dramatically reduces forest carbon stocks. For example, in the tropics, where big trees occur in low densities, and logging removes only a few big trees per hectare, forest carbon decreases by an average of 35% after just one cut and takes about a century to fully recover [44].

Figure 13: An Example of an emergent tree crown, way above the canopy from a Mexican rainforest. Such 'big trees' are present in primary forests of nearly all types, with the exception of cold boreal forests. Photo credit: Jaime Rojo.

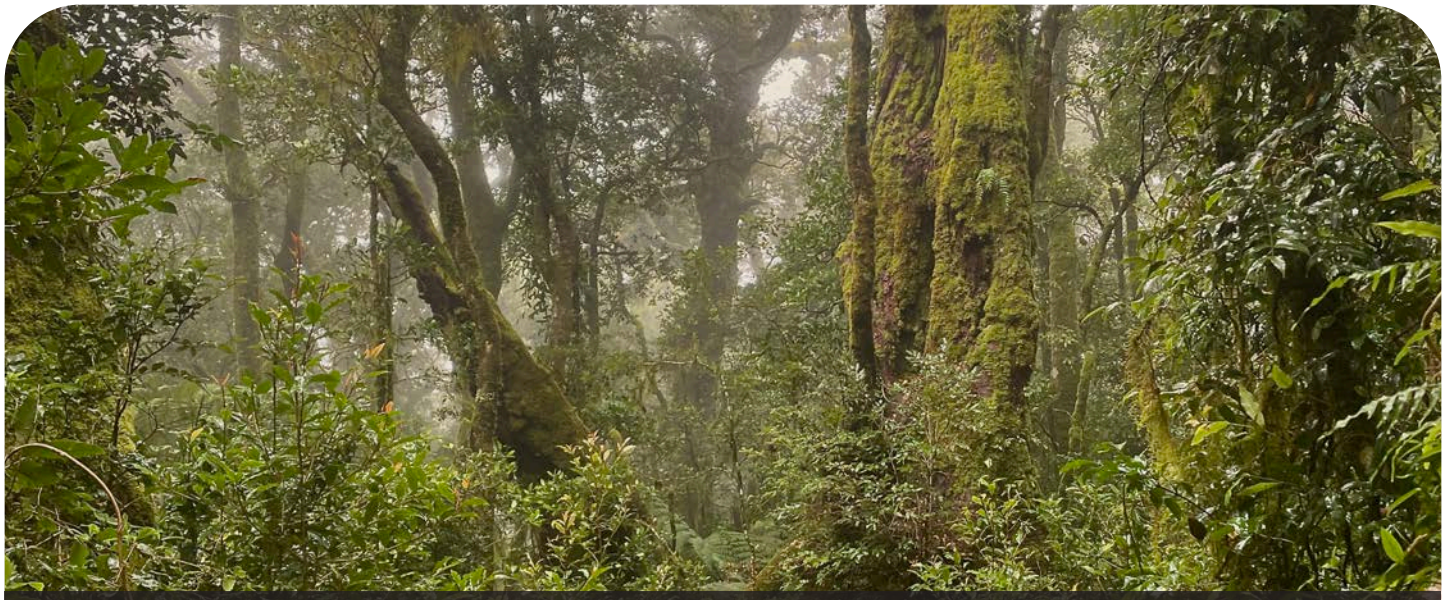


Figure 14: Antarctic Beech (*Lophozonia moorei*) forests of the Gondwana Rainforests World Heritage Area, Australia. Through natural copicing and resprouting specimens can age to more than two thousand years. Credit: A. Buckwell.

6. Understanding Climate–Biodiversity synergies

Biodiversity and ecosystem integrity underpin ecosystem services, including climate change mitigation.

Forest ecosystems with higher ecosystem integrity provide better climate mitigation as they store more carbon and also store that carbon more securely into the long term (figure 14). In other words, the carbon stocks in the forest or other ecosystem are at lower risk of being reemitted [45]. Conversely, degrading (or clearing ecosystems) emits carbon dioxide and weakens ecosystems, decreasing their resilience [46].

Therefore, integrating climate action with protecting and restoring biodiversity results in a mutually reinforcing, virtuous cycle, where both are strengthened, together. Alternatively, ignoring this interdependence can result in a downward mutually reinforcing spiral, where degrading or clearing ecosystems causes losses in biodiversity, as well as more emissions, worsening climate change and worsening the outlook for biodiversity.

Common misperceptions about climate biodiversity synergies are:

1. Climate - biodiversity synergies do not imply that more biodiverse ecosystems store more carbon.

Primary tropical forests are the most diverse terrestrial ecosystems on the planet, and they do store a lot of carbon in their vegetation because they are very productive ecosystems and have very dense vegetation [47].

However, unless they are located on peatlands, they store relatively less carbon in soils when compared to temperate and boreal biomes [48]. Temperate and boreal forests store less carbon in trees as they experience a much shorter growing season. Therefore, it is the climate as well as the evolved biodiversity that determines the total carbon content of a forest - all forests (indeed, all ecosystems) need their natural complement of biodiversity to maximise their carbon storage [49].

2. Climate - biodiversity synergies refer to benefits to biodiversity if climate change impacts are reduced and the benefits to the climate if biodiversity and ecosystem integrity are protected and restored.

While biodiversity clearly benefits if we prevent catastrophic climate impacts, so too the climate benefits if we retain and recover long-term ecosystem carbon stocks - which, to great extent, depends on protecting and restoring biodiversity. Therefore, climate and biodiversity synergies are mutually reinforcing [50].

7. The way forward

The world has lost one third of its forests, and only a quarter of what remains is in good ecological condition. With the conclusion of the **United Nations Strategic Plan for Forests 2017-2030** just four years away, deforestation and degradation are still driving biodiversity loss and pushing emissions higher. Tree-cover statistics mask the crisis by not correctly accounting for deforestation and degraded forests, and for the different types of forests, such as plantations. Protecting primary forests, along with other high integrity forest ecosystems, is the most cost-effective strategy for maximising biodiversity and climate outcomes and meeting 2030 global goals.

Primary forests are the most biodiverse terrestrial ecosystems on Earth. Their native species, structure and big, old trees make them resilient to fire, drought and disease, and allow them to hold vast carbon stocks securely for centuries (figure 15). Logging even a few large trees per hectare can cut carbon by a third and take a century to recover. Carbon lost from these forests cannot be rebuilt in time to meet our climate goals.

The boundaries of intact forest landscapes and the networks of smaller areas of remnant primary forest also serve as foci and 'anchors' for ecological restoration and landscape connectivity initiatives. And recognizing and respecting the territorial rights of their Indigenous customary stewards, and providing them with the necessary capacity and resources, has also proven to be a pre-requisite for sustainable outcomes.

Climate stability and biodiversity conservation are therefore one challenge with one solution.



Figure 15: Primary forests are the most biodiverse terrestrial ecosystems on Earth. Their native species, structure and big, old trees make them resilient to fire, drought and disease, and allow them to hold vast carbon stocks securely for centuries. Photo credit: Frederic Demeuse-Bogachiel.

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